

Alyssa Rivera

(619) 739 - 1445 • San Diego, CA • alyssagarivera@gmail.com • www.linkedin.com/in/alygrivera
U.S. Citizen | Eligible to Obtain U.S. Government Security Clearance

EDUCATION

San Diego State University | B.A. Applied Arts and Sciences, Computer Science

May 2025

SKILLS SUMMARY

- Programming Languages: Python, C++, C#, Java, SQL, JavaScript, CSS, R, HTML, Shell Scripting, HLSL
- Software Development: Object-oriented programming, software architecture, Git version control, database design, unit testing, application deployment, agile development, requirement analysis
- Frameworks & Technologies: React, .NET, Flask, Ubuntu, Vue.js, MongoDB, PostgreSQL, RESTful APIs, TensorFlow, scikit-learn, WPF
- Tools & Platforms: Git, GitHub, Visual Studio, VS Code, Active Directory
- Cloud & Certifications: Google Cloud Data Analytics Certificate, AWS Glue, Amazon Athena

SOFTWARE ENGINEERING PROJECTS

Frozen Assets, 2024 (Web-Based Application)

Served as a Test Manager for a full-stack inventory and logistics management platform developed using Python, Flask, SQL, JavaScript, Vue.js, Bootstrap, and GitHub. The system integrated order entry, inventory management, shipment tracking, invoicing, and trouble-ticket management into a modular architecture supporting role-based access control, inventory synchronization, and transactional consistency. Led QA planning, integration testing, risk assessment, and JAD documentation efforts within a six person development team.

- Contributed to the development of a web-based inventory and logistics management platform by collaborating in a six person team using Python, Flask, SQL, JavaScript, Vue.js, and GitHub, producing a modular application that supports order management, shipment tracking, invoicing, and customer support functionality.
- Defined system requirements and application specifications by authoring a Joint Application Design (JAD) document, establishing a strong foundation for team coordination and application development.
- Designed application and subsystem interactions by analyzing requirements and documenting data flow between inventory, order entry, shipment tracking, and ticket management components, improving maintainability and system organization.
- Implemented and validated backend functionality by utilizing SQL databases, API integrations, and transactions, ensuring reliable data processing and consistency within the system.
- Supported iterative development and deployment activities by coordinating both testing efforts and release validation, improving application stability throughout the project lifecycle.
- Collaborated with team members by reviewing requirements, validating functionality, and defining system behavior, contributing to successful project delivery according to requirements.

Tea Steeper Agent, 2025

Developed an embedded tea-steeping assistant in C++ on a Raspberry Pi using a DS18B20 temperature sensor to automate temperature monitoring and steep-time management. The system allowed users to select tea type and strength, dynamically calculating target temperatures and steep durations while continuously monitoring live sensor data until brewing conditions were met. Built using object-oriented design, state machine architecture, and Makefile build automation, the project emphasized hardware/software integration, sensor validation, and embedded systems development through iterative testing and debugging.

- Developed an embedded tea-steeping assistant by implementing object-oriented C++ software on a Raspberry Pi, enabling automated temperature monitoring and steep-time tracking.
- Designed system architecture by creating a state-machine framework that manages user input, monitors temperature, conducts timing operations, and notifies users.
- Integrated DS18B20 temperature sensors by applying hardware communication and threshold detection logic, enabling environmental monitoring in real time.
- Implemented configurable brewing operations by developing algorithms that dynamically calculate target temperatures and steep duration based on selections made by users.
- Automated build and deployment proceeds by creating Makefile compilations, reducing development overhead and

improving project maintainability.

- Validated software functionality through iterative testing and debugging, ensuring reliable operation across multiple brewing scenarios.

Cubelet, 2024

Served as a level designer on a five person team developing a puzzle-horror game in Unity using C#, Rigidbody physics, ScriptableObjects, coroutines, Cinemachine, and the Unity Input System. The project combined physics-based puzzle solving with progressive mechanic unlocks, requiring players to manipulate environmental objects, activate pressure plates, and navigate increasingly complex challenges before transitioning into a horror experience featuring flashlight-based survival mechanics and enemy AI. The project emphasized gameplay progression, puzzle sequencing, and player interactions.

- Developed gameplay systems for a puzzle horror game in the Unity game engine by implementing mechanics using C#, Rigidbody physics, ScriptableObjects, and coroutines, contributing to a complete player experience.
- Designed level architecture and gameplay progression by creating interconnected puzzle mechanics, environmental interactions, and objective-driven challenges.
- Implemented physics based interaction systems by utilizing Unity Rigidbody and collision systems, enabling object manipulation and problem solving.
- Integrated modular gameplay components by using ScriptableObjects, coroutines, and Cinemachine, improving code organization and system maintainability.
- Collaborated in a five person development team by refining gameplay requirements, iterating on feature implementation, and supporting project delivery.
- Performed debugging and system optimization by resolving interaction, collision, and physics related issues, improving gameplay stability and user experience.

Titanic Survivor Prediction, 2024

Developed a machine learning classification system in Python using NumPy and pandas to predict passenger survival outcomes from the RMS Titanic dataset. The project analyzed historical passenger records sourced from the Kaggle Titanic dataset to evaluate relationships between demographic, socio-economic, and environmental factors and their influence on survival probability. Through data preprocessing, feature analysis, and comparative model evaluation, the project assessed the predictive performance of logistic regression, decision tree, and gradient descent algorithms.

- Developed a machine learning application by implementing data processing and classification pipelines using Python, NumPy, and pandas, enabling prediction of passenger survival outcomes from historical datasets.
- Prepared data by cleaning, transforming, and validating raw passenger records, improving data quality and model reliability.
- Implemented and evaluated machine learning algorithms by developing logistic regression, decision tree, and gradient descent models, identifying the decision tree as the highest performing model at 91.92% accuracy.
- Engineered reusable data analysis processes by automating preprocessing and model evaluation tasks, improving development efficiency and reproducibility.
- Analyzed feature relationships and predictive performance by applying statistical and machine learning techniques, generating valuable insight from large datasets.
- Validating model performance through structured testing and iterative refinement, improving prediction accuracy and reducing model noise.

WORK EXPERIENCE

Information Technology - Automation

May 2026 - Present

San Diego Police Department | San Diego, CA

Support technology operations, device lifecycle management, and process automation initiatives for the San Diego Police Department using PowerShell, WPF, SQL, Active Directory, and enterprise asset-management tools.

- Developed enhancements for a PowerShell-based maintenance application by implementing WPF user interface components, progress tracking functionality, and user configuration options, improving usability and efficiency.
- Designed and implemented automation operations by utilizing PowerShell scripting and Active Directory integration,

reducing manual effort in maintenance processes.

- Leveraged SQL databases and inventory management systems by creating data retrieval and validation behaviors, improving asset tracking accuracy and operational visibility.
- Participated in technology deployment initiatives by validating hardware functionality and supporting device provisioning processes, ensuring operational readiness.
- Authored technical documentation and process specifications by documenting maintenance and application procedures, improving knowledge transfer and maintainability.

Barista

August 2025 - Present

Starbucks | San Diego, CA

- Deliver exceptional customer service in a high-volume environment using strong communication, adaptability and problem-solving skills.
- Manage multiple concurrent tasks under intense situations, changing priorities to maintain speed and accuracy during peak hours. Adhere to strict standardized procedures, ensuring consistency in quality and detail.

Teaching Assistant - 3D Game Programming

July 2024 - June 2025

San Diego State University, Computer Science Department | San Diego, CA

Served as a Teaching Assistant for a 3D game programming course centered on software engineering principles within interactive system development using the Unity game engine and C# in Visual Studio Code.

- Mentored 48 students per semester by providing technical guidance on software engineering principles, object-oriented programming, and application development within Unity and C# environments.
- Assisted students in designing and implementing gameplay systems by reviewing software architecture, modular design patterns, and system integration strategies, improving project quality and maintainability.
- Debugged complex software issues by utilizing Visual Studio debugging tools and structured troubleshooting methodologies, helping students resolve runtime failures and performance bottlenecks.
- Developed instructional resources and reusable coded examples by demonstrating scalable software engineering practices improving student understanding of maintainable code design.
- Guided students through version control and collaborative development processes by supporting GitHub repository management and team based software development practices.

Lead Coordinator & Player - Aztec Gaming

May 2023 - July 2025

San Diego State University | San Diego, CA

- Lead event and tournament organization for over 100 members of the collegiate Valorant Esports division.
- Member of the Overwatch 2 Esports team, contributing to strategy development and in-game decision making.
- Analyzed team compositions and player feedback to improve match balance and team strategies.

Development Assistant

September 2021 - August 2025

KPBS | San Diego, CA

- Organize and support 10+ events annually for over 2,500 major-giving KPBS members.
- Maintain and update member databases with 100% data accuracy.
- Manage financial transactions and reimbursements with a focus on precision and confidentiality.